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# The *C. elegans* gustatory receptor homolog LITE-1 is a chemoreceptor required for diacetyl avoidance

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**eLife Assessment**

Avoidance of UV and blue light by the nematode C. elegans is mediated by the unusual transmembrane protein LITE-1, a non-canonical photoreceptor. In this **valuable** work, the authors provide **convincing** evidence that LITE-1 function is also required for avoidance of very high concentrations of the food-associated cue diacetyl, suggesting that it may also function as a diacetyl chemoreceptor. While the evidence for this idea is **incomplete**, these intriguing findings suggest an unexpected complexity in the function of this unusual photoreceptor.

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## Abstract

The nematode *C. elegans* does not have eyes but can respond to aversive UV and blue light stimulation and even distinguish colours. The gustatory receptor homolog LITE-1 was identified in forward genetic screens for worms that failed to respond to blue light stimulation. When LITE-1 is expressed in body-wall muscles, it causes contraction in response to blue light suggesting that LITE-1 is both necessary and sufficient for blue light response. Here we show that in addition to light avoidance, LITE-1 is also required for worms' avoidance of high concentrations of diacetyl, an odorant that is attractive at low concentrations. Like blue light, diacetyl causes muscle contraction in transgenic worms engineered to express LITE-1 in body-wall muscles. These data are consistent with a direct chemoreceptor function for LITE-1 which would make it a multimodal sensor of aversive stimuli.

# Introduction

Sensory perception is fundamental to how organisms interact with their surroundings. Animals, including *Caenorhabditis elegans*, detect and respond to a wide range of environmental cues such as light and odorants. Despite having a compact nervous system of only 302 neurons, *C. elegans* exhibits sensory capabilities that allow it to effectively navigate and adapt to its environment (Bargmann, 2006; Cook et al., 2019). Odorants such as volatile organic compounds are detected via specialised olfactory receptor neurons. These neurons express specific receptors that bind to specific molecules and trigger distinct behavioural responses (Bargmann, 2006; Bargmann et al., 1993; Bargmann & Horvitz, 1991). Diacetyl, a volatile organic α-diketone found in *C. elegans*' natural environment elicits opposing behaviours depending on its concentration. At low concentrations, diacetyl is sensed by the olfactory receptor ODR-10, promoting attraction (Itskovits et al., 2018; Sengupta et al., 1996; Zhang et al., 1997). In contrast, higher concentrations induce an avoidance response, thought to be mediated by the receptor SRI-14 (Taniguchi et al., 2014).

In addition to chemical cues, *C. elegans* also detects and responds to light, despite lacking eyes or classical photoreceptors. This light sensitivity is mediated by LITE-1, a member of the Gustatory Receptor family (Montell, 2009). LITE-1 was first identified through genetic screens for mutants defective in light avoidance behaviour (Edwards et al., 2008). Unlike canonical photoreceptors such as rhodopsins or opsins, LITE-1 shares little sequence similarity with known light sensing proteins (Edwards et al., 2008; Gong et al., 2016; Liu et al., 2010). Further underscoring its uncanonical nature, LITE-1 adopts an inverted membrane topology that is unlike classical G protein-coupled receptors (GPCRs) (Gong et al., 2016).

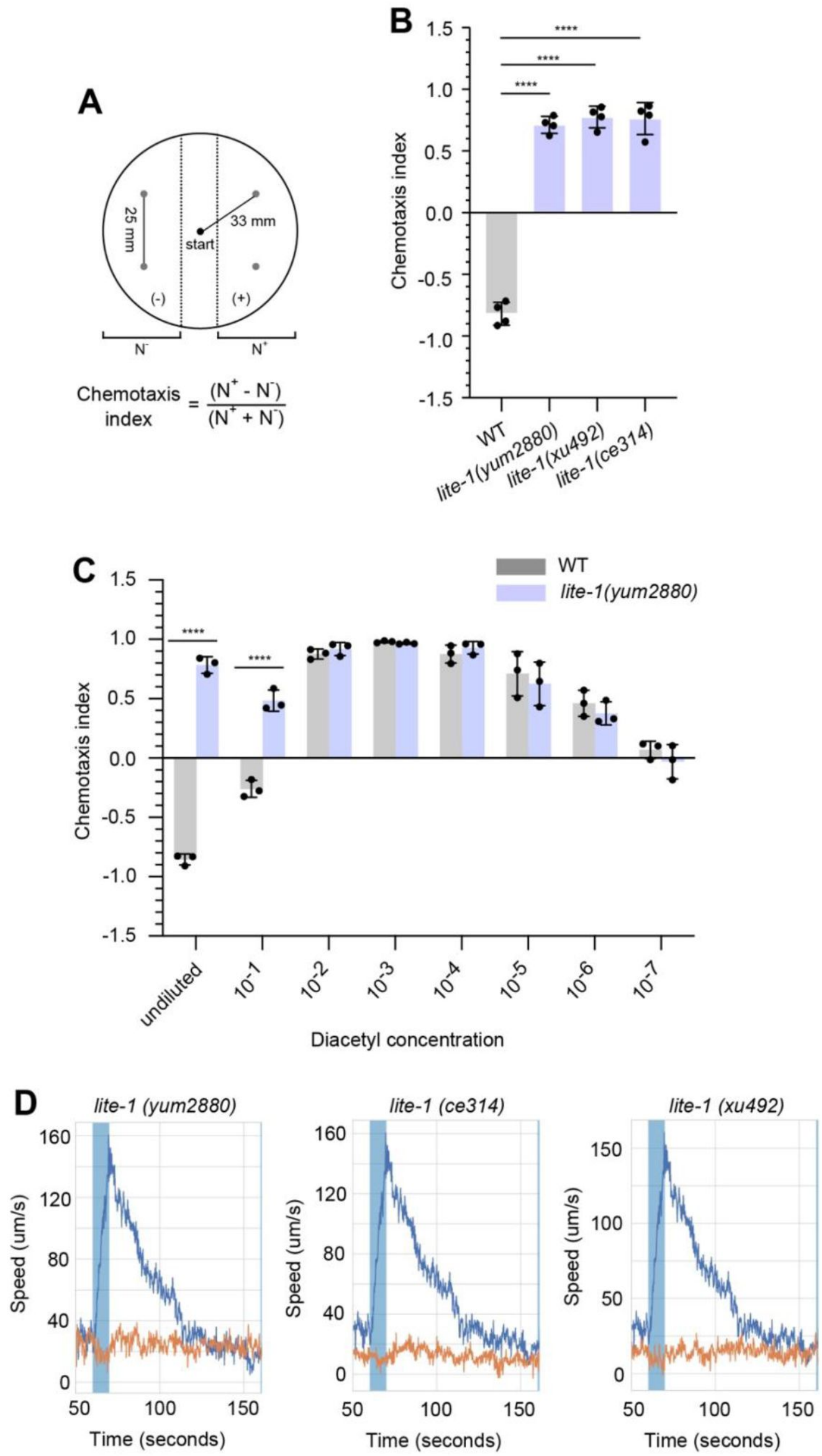
LITE-1 phototaxis behaviour is driven by multiple light responsive sensory neurons, including ASJ, ASH and ASK (Liu et al., 2010; Ward et al., 2008; Zhang et al., 2022). Signal transduction occurs via G protein signalling, upregulation of cGMP levels and activation of cyclic nucleotide-gated (CNG) channels, leading to a rapid avoidance response (Edwards et al., 2008; Liu et al., 2010; Ward et al., 2008). Recent studies suggest that LITE-1 may function as a light activated ion channel with a putative ligand binding pocket that is activated by ultraviolet and blue light via key tryptophan residues (Edwards et al., 2008; Gong et al., 2016; Hanson, Scholüke, et al., 2023; Ward et al., 2008).

Here we show that beyond its role in photoreception, LITE-1 is also required for the avoidance of high concentrations of diacetyl and that diacetyl induces muscle contraction in transgenic worms expressing LITE-1 in body-wall muscles. Our findings support the notion that LITE-1 functions as a multifunctional receptor involved in both light and chemical sensing.

# Results

## LITE-1 is required for diacetyl and blue light avoidance

A recent screen of a panel of worm strains that collectively contained knockouts of all non-essential GPCRs identified a triple mutant that failed to avoid high concentrations of diacetyl (Pu et al., 2023). One of the mutated genes was *lite-1*. We performed a chemotaxis assay to test single mutants of several loss-of-function alleles of *lite-1* and found they all failed to avoid high concentrations of diacetyl (**Fig. 1A** and **1B**) while remaining attracted to lower concentrations (**Fig. 1C**). As expected, these *lite-1* mutants no longer respond to blue light stimulation (**Fig. 1D**) (Edwards et al., 2008).




## Figure 1

(A) Schematic of chemotaxis assay setup. (B) Chemotaxis index of wild-type animals and mutants with different alleles of *lite-1* that have defects in diacetyl response. P-values are from unpaired two-tailed t-tests, n = 4 biological replicates, each an average of 4 plates. (C) Chemotaxis index of wild-type and *lite-1 (yum2880)* animals with different diacetyl concentrations (5 µl). P-values are from unpaired two-tailed t-tests, n = 3 biological replicates, each an average of 4 plates. (D) Time series of wild-type (blue line) animals and mutants with different alleles of *lite-1* (orange line) when stimulated with blue light (blue shaded box). n = 3 biological replicates, each an average of 12 wells.

### Chemosensory neurons ADL, ASK and ASH are involved in avoidance of diacetyl

To identify neurons involved in the response to high concentrations of diacetyl, we used calcium imaging to record from 11 pairs of sensory neurons in worms exposed to a pulse of diacetyl (**Fig. 2A**). In *lite-1* mutants, the LITE-1-expressing sensory neurons ADL and ASK showed a clear defect in diacetyl response, whereas another LITE-1-expressing neuron ASH showed a normal response (compared to WT animals, **Fig. 2B**). The response of the other tested neurons was not significantly different between mutant and WT animals (**Fig. 2B**). The normal response to diacetyl in ASH in the absence of LITE-1 is likely due to its expression of SRI-14 which has previously been shown to respond weakly to high concentration of diacetyl (Taniguchi et al., 2014). We also recorded from *unc-13* mutants which are impaired in synaptic activity and found little effect on the ADL and ASK responses (**Fig. 2C**) (Richmond et al., 1999). This is consistent with direct sensation in these neurons although neural signalling through gap junctions or via neuromodulation by other receptors could not be ruled out.

### LITE-1 is sufficient for diacetyl response in other cell types

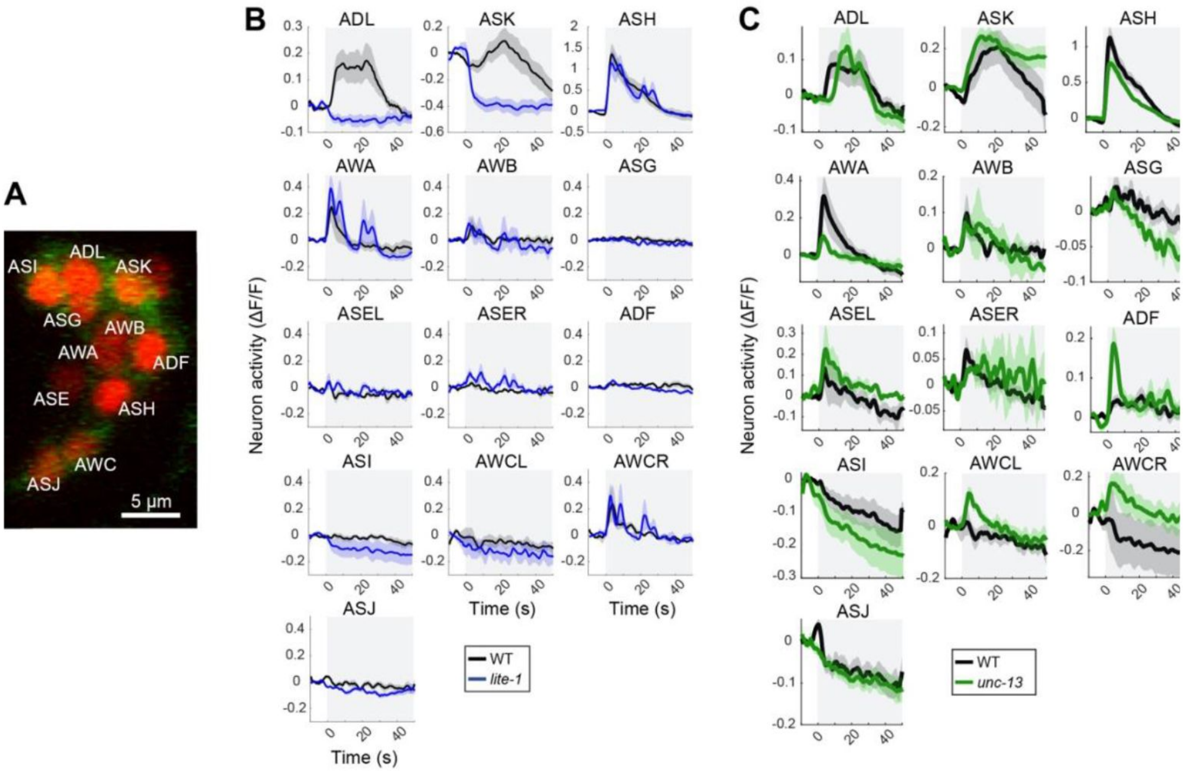
To test whether LITE-1 is sufficient for a response to high diacetyl concentrations, we used a transgenic strain that expresses LITE-1 in body-wall muscles (Edwards et al., 2008). Worms ectopically expressing LITE-1 in body muscle cells evoke muscle contraction and shortening of body length in response to blue light stimulation (Edwards et al., 2008). When these worms are exposed to a 1:50 dilution of diacetyl, they showed a rapid and near complete paralysis whereas wild-type animals continue to swim (**Fig. 3A** and **3B**). These worms also showed significant contraction in average body length, contracting by ~11% after diacetyl exposure (**Fig. 3C**). These results suggest that LITE-1 is directly activated by diacetyl although it remains possible that it is a diacetyl metabolite that activates LITE-1 given that the full contraction develops over a minute time scale. Consistent with a ligand receptor relationship, molecular docking predicts low micromolar affinity of diacetyl binding to LITE-1 in the same binding pocket that was recently identified as a putative chromophore binding site (**Fig. 3D**) (Hanson, Scholüke, et al., 2023).

### 2,3-pentanedione also interacts with LITE-1

We next tested whether *lite-1* is required for avoidance of other high concentration odorants. Of the seven odorants tested, *lite-1* mutants showed a defective response to 2,3-pentanedione, a chemical structurally related to diacetyl (**Fig. 4A**). The effect was weaker than that observed for diacetyl, with worms still avoiding high concentrations of 2,3-pentanedione but more weakly than wild-type animals. Supporting this observation, molecular docking predicts interaction between 2,3-pentanedione and LITE-1 putative binding pocket (**Fig. 4B**). Like diacetyl, 2,3-pentanedione also evokes paralysis and muscle contraction of worms expressing LITE-1 in muscle cells (**Fig. 4C** and **4D**).

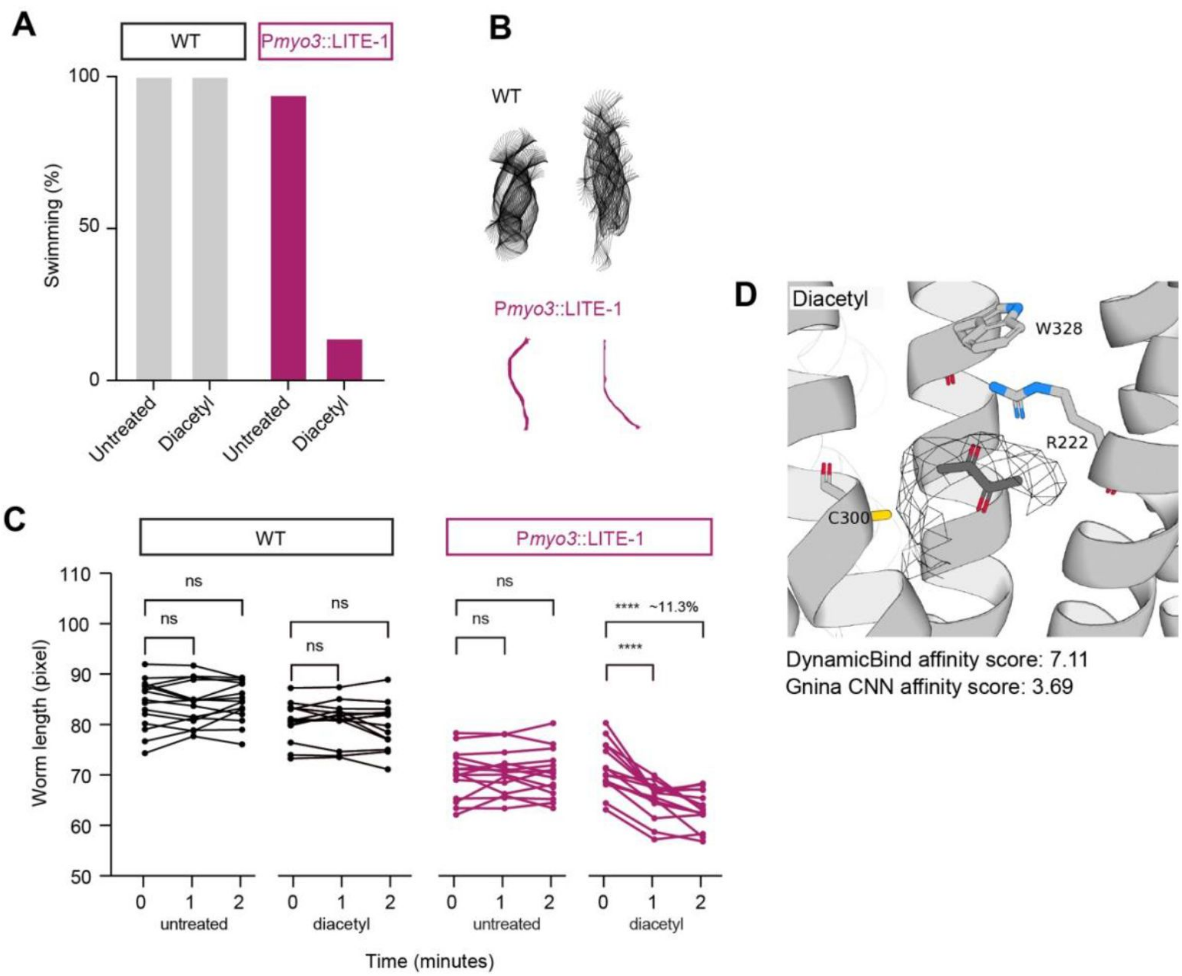
## Discussion

LITE-1 is best known for its role in light sensation in *C. elegans*, but how it senses light is not yet clear. In particular, an indirect mechanism in which light produces photoproducts that activate the channel has not been ruled out and *a priori* has always seemed plausible given that LITE-1 is similar to gustatory receptors that are known to sense chemicals (Montell, 2009). A direct diacetyl response would make a photoproduct-based mechanism more plausible. The identification of a second structurally related compound, 2,3-pentanedione may help in narrowing



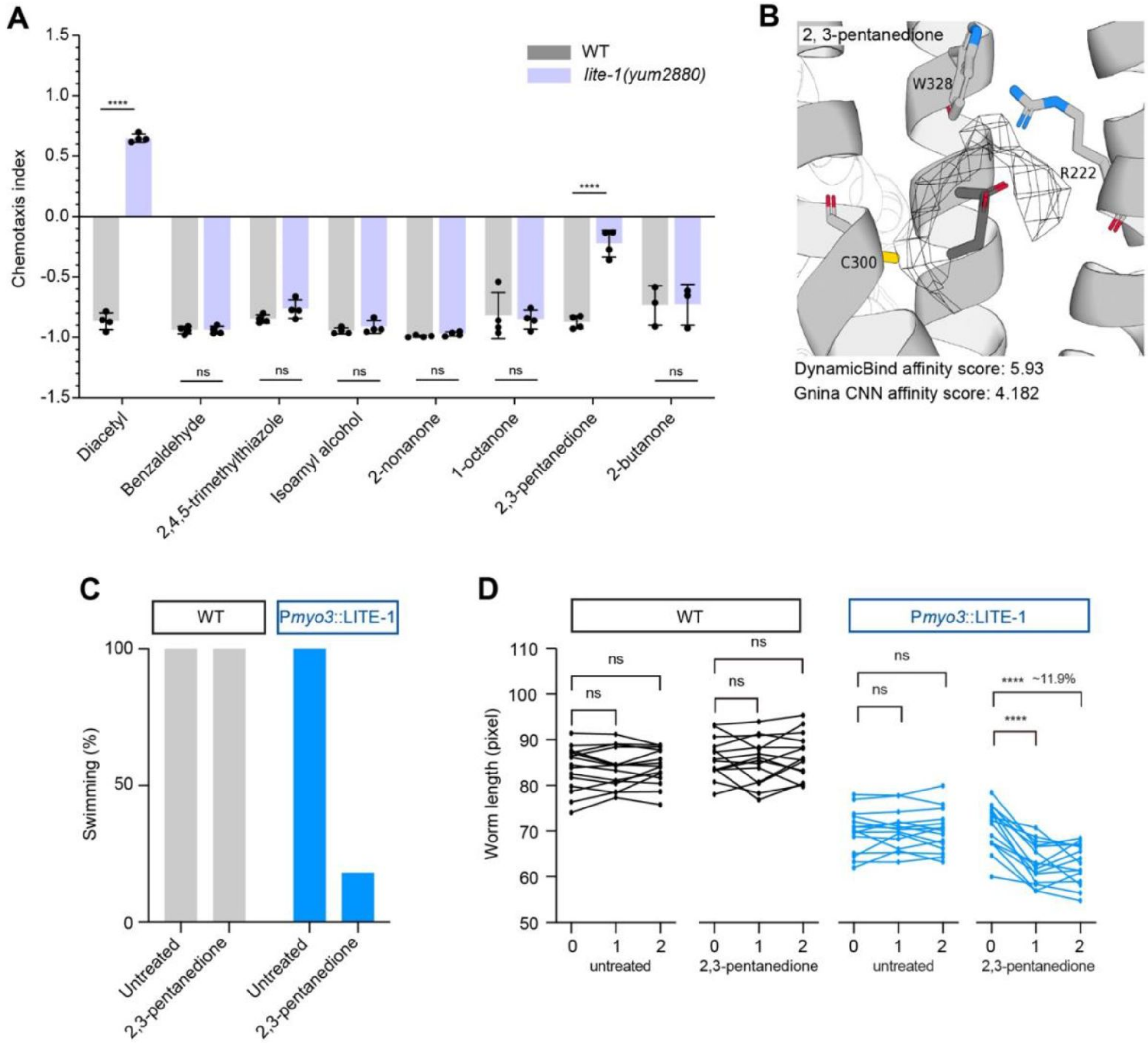

## Figure 2

(A) Confocal image of labelled sensory neurons and calcium imaging traces showing the responses of three neurons when exposed to high concentrations of diacetyl. (B) Neuronal activity of wild-type controls with *lite-1* mutants. (C) Neuronal activity of wild-type controls to *unc-13* mutants. Grey shaded box denotes the presence of diacetyl. Shaded area around the activity dynamics denotes standard error.




## Figure 3

(A) Percent of worms that are swimming at two minutes after exposure to buffer untreated or to diacetyl. Wild-type animals continue swimming in buffer and diacetyl (WT, left) whereas worms expressing LITE-1 in body wall muscles (*Pmyo-3::*LITE-1, right) are mostly paralysed when treated with diacetyl. (B) Sample midline trajectories from wild-type and *Pmyo-3::*LITE-1 animals showing difference in motion (each trajectory shows 6 seconds of swimming). (C) Worm body length over time after treatment. Wild-type animals do not show a significant length difference in either buffer or diacetyl treatments (left) whereas worms expressing LITE-1 in body wall muscles show a significant contraction in response to diacetyl (right; insert showing the average contraction). P-values from paired two-tailed t-test, n ≥ 12 worms. (D) Molecular docking of diacetyl into the AlphaFold-predicted structure of LITE-1 with predicted micromolar affinity.




## Figure 4

(A) Chemotaxis index of wild-type animals and *lite-1* mutants for known odorants. P-values are from unpaired two-tailed t-tests, n = 4 biological replicates each an average of 4 plates. (B) Molecular docking of 2,3-pentanedione into the AlphaFold-predicted structure of LITE-1 with predicted micromolar affinity. (C) Percent of worms that are swimming at two minutes after exposure to buffer untreated or to 2,3-pentanedione. Wild-type animals continue swimming in buffer and 2,3-pentanedione (WT, left) whereas worms expressing LITE-1 in body wall muscles (*Pmyo-3::*LITE-1, right) are mostly paralysed when treated with 2,3-pentanedione. (D) Worm body length over time after treatment. Wild-type animals do not show a significant length difference in either buffer or 2,3-pentanedione treatments (left) whereas worms expressing LITE-1 in body wall muscles show a significant contraction in response to 2,3-pentanedione (right; insert showing the average contraction). P-values from paired two-tailed t-test, n ≥ 12 worms.

down photoproducts with similar moieties. Direct and photoproduct light sensing are not mutually exclusive. Based on LITE-1's predicted structure and its absorption spectrum, it has recently been proposed that UV may be directly sensed while the blue light response may be mediated through a photoproduct or an unidentified chromophore (Hanson et al., 2023).

While our study primarily focused on chemosensory neurons, it remains possible that non-chemosensory neurons expressing LITE-1 may contribute to diacetyl detection or modulating the behavioural response. Indeed, a recent study demonstrated that the LITE-1-expressing interneuron AVG response to blue light stimulation by releasing the neuropeptide NLP-10 to mediate the escape behaviour (Dunkel et al., 2025). Nevertheless, phototransduction and chemotransduction appear to involve shared sensory neurons, such as ASK and ASH, suggesting a potential overlap in their underlaying signalling mechanisms (Zhang et al., 2022). It is possible that LITE-1 chemosensory response may rely on components necessary for LITE-1 photosensory function, such as G proteins (GOA-1, GPA-3), guanylate cyclase (DAF-11, ODR-1), CNG channels (TAX-2,4) and phosphodiesterase (PDE-1,2,5) (Liu et al., 2010; Ward et al., 2008). Furthermore, the use of *lite-1* point mutants that affect specific LITE-1 function such as light sensing, channel gating or binding pocket could further elucidate LITE-1 mechanisms (Gong et al., 2016; Hanson, Scholüke, et al., 2023).

We have shown that LITE-1 is required for the avoidance of high concentrations of diacetyl and a related compound 2,3-pentanedione. Furthermore, we have shown that LITE-1 is sufficient to make muscle cells responsive to diacetyl and 2,3-pentanedione, consistent with a direct response. Diacetyl is an ecologically relevant molecule present in *C. elegans*' natural environment that has been hypothesised to be involved in prey-finding since low concentrations produced by lactic acid bacteria on rotting fruit is attractive (Choi et al., 2016). Avoidance of high concentrations of diacetyl may therefore also be ecologically relevant. In this case, LITE-1 has likely evolved to be a multimodal receptor of aversive stimuli akin to other multimodal receptors such as TRPV1, which responds to heat, capsaicin, and acidic pH (Zhang et al., 2023).

# Material and methods

## Bacterial and worm strains maintenance

The *E. coil* wild-type isolate OP50 was sourced from Caenorhabditis Genetics Center (CGC) and maintained on lysogeny broth (LB) agar under standard laboratory conditions. The following worm strains were used in this study, N2; wild-type isolate (CGC), CHS1610; *lite-1(yum2880)* (Pu et al., 2023), KG1180; *lite-1(ce314)* (Edwards et al., 2008), KG1272; *cels37[myo-3$_{prom}$::lite-1::GFP]* (Edwards et al., 2008), TQ8245; *lite-1(xu492)* (Zhang et al., 2020), ZAS280; *In[osm-6::GCaMP3, osm-6::ceNLS-mCherry-2xSV40NLS]* (Iwanir et al., 2019), ZAS371; *unc-13(s69) In[osm-6::GCaMP3, osm-6::ceNLS-mCherry-2xSV40NLS]* (Bokman, Kalij, et al., 2024; Bokman, Pritz, et al., 2024), ZAS526; *lite-1(ce314) In[osm-6::GCaMP3, osm-6::ceNLS-mCherry-2xSV40NLS]*, were cultured on Nematode Growth Medium seeded with *E. coli* (OP50) as previously described (Brenner, 1974). Worm synchronization was performed by bleaching unsynchronised gravid adults (Barlow, 2019).

## Chemotaxis assay

The chemotaxis assays were conducted at 20°C in 9 cm plates filled with 10 ml of agar (1.7% agar, 1 mM $CaCl_2$, 1 mM $MgSO_4$ and 25 mM $K_2HPO_4$ pH 6). Two parallel lines, 1 cm apart, were drawn on the middle of the plates which define the area to place the animals. Two odorant spots were designated on one side of the plate, while two control spots were positioned on the opposite side (**Fig. 1A**). 1 µl of 1 M $NaN_3$ was added to all four spots to immobilise the animals. Different volumes of each odorant: diacetyl (5 µl), 2,4,5-trimethylthiazole (7.5 µl), benzaldehyde (2.5 µl), isoamyl alcohol (7.5 µl), 2,3-pentanedione (10 µl), 2-nonanone (1 µl), 1-octanol (2.5 µl), 2-butanone

(10 μl) and equal volume of ethanol (control) were spotted onto the respective dots. Synchronized day-one adults were washed 3 times in chemotaxis buffer (1 mM $CaCl_2$, 1 mM $MgSO_4$ and 25 mM $K_2HPO_4$ pH 6) before being transferred onto the centre of the assay plate where the assay will last for 1 hr at 20 °C. The chemotaxis plates were then stored at 4 °C before counting. The chemotaxis index was calculated as (number of animals on the odorant side – number of animals on the ethanol side) / (sum of all animals on both sides).

## Paralysis assay

The paralysis assays were conducted at 20°C in 6-well plates with 4 ml of agar (1.7% agar, 1 mM $CaCl_2$, 1 mM $MgSO_4$ and 25 mM $K_2HPO_4$ pH 6). Synchronized day-one adults were washed 3 times in chemotaxis buffer (1 mM $CaCl_2$, 1 mM $MgSO_4$ and 25 mM $K_2HPO_4$ pH 6) before being transferred onto the centre of the 6-well plate. 1 ml of chemotaxis buffer containing diacetyl (1:50) was then added into each well and videos were immediately acquired at 25 frames per second using a shutter speed of 25 ms and a resolution of 12.4 μm/pixel.

## Blue light stimulation assay

The blue light stimulation assays were conducted at 20°C in 24-well plates containing agar (1.7% agar, 1 mM $CaCl_2$, 1 mM $MgSO_4$ and 25 mM $K_2HPO_4$ pH 6). Synchronized day-one adults were transferred onto the centre of the 24-well plate pre-seeded with OP50. The plates were then transferred onto the multi-camera tracker for 30 min to allow acclimatization to the tracker room before imaging. Videos were acquired at 25 frames per second using a shutter speed of 25 ms and a resolution of 12.4 μm/pixel. Videos were taken in the following order: 300 s pre-stimulus, blue light stimulus with one 10 s pulse of blue light at 60 s and a 100 s post-stimulus recording. Videos were then processed with Tierpsy Tracker as previously described (Barlow et al., 2022; Javer et al., 2018).

## Calcium imaging

Young adult worms were inserted into a microfluidic chip and anesthetized with 10 mM levamisole and allowed to habituate for 10 mins as previously described (Chronis et al., 2007). Recordings began with 1 min of light habituation, after which the stimulus was presented for 1 min The stimulus used was 1:5 diacetyl in CTX buffer (5 mM $KH_2PO_4/K_2HPO_4$ pH 6, 1 mM $CaCl_2$ and 1 mM $MgSO_4$). Recordings were made using a Nikon A1R + confocal laser scanning microscope with a water immersion × 40 (1.15NA) objective controlled by the Nikon NIS-elements software. Z-axis resolution was 0.5-0.8 μm. Acquisition rate was ~1.5 volumes/second, interpolated to a final framerate of 2 Hz.

## Neuron identification and signal extraction

Neuron segmentation and tracking was done on the nuclear mCherry signal using an algorithm developed by (Toyoshima et al., 2016). Neurons were identified visually based on anatomical positions. Neurons that could not be unambiguously identified were removed from analysis. Calcium readings were taken from within 0.9 of the radius of the segmented nuclear sphere. Signal intensity was normalized by baseline activity, defined as the lowest point of a 10-frame running average. All data analyses were performed using in-house MATLAB scripts (Pritz et al., 2023).

## Molecular docking

Molecular docking was performed on the LITE-1 tetramer structure using DynamicBind (Hanson, Scholuke, et al., 2023; Lu et al., 2024). Additional rounds of docking on top ranked poses were performed with Gnina (McNutt et al., 2021).

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# Additional information


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# References


Bargmann (2006) **Chemosensation in C. elegans** *WormBook* https://doi.org/10.1895/wormbook.1.123.1 | Google Scholar

Bargmann C. I., Hartwieg E., Horvitz H. R. (1993) **Odorant-selective genes and neurons mediate olfaction in C. elegans** *Cell* **74**:515–527 https://doi.org/10.1016/0092-8674(93)80053-h | Google Scholar

Bargmann C. I., Horvitz H. R. (1991) **Chemosensory neurons with overlapping functions direct chemotaxis to multiple chemicals in C. elegans** *Neuron* **7**:729–742 https://doi.org/10.1016/0896-6273(91)90276-6 | Google Scholar

Barlow I. (2019) **Bleach Synchronisation of C. elegans v1** *protocols.io* https://doi.org/10.17504/protocols.io.2bzgap6 | Google Scholar

Barlow I. L., Feriani L., Minga E., McDermott-Rouse A., O'Brien T. J., Liu Z., Hofbauer M., Stowers J. R., Andersen E. C., Ding S. S., Brown A. E. X. (2022) **Megapixel camera arrays enable high-resolution animal tracking in multiwell plates** *Commun Biol* **5**:253 https://doi.org/10.1038/s42003-022-03206-1 | Google Scholar

Bokman E., Kalij I. P., Zaslaver A. (2024) **Aberrant Positions of the Chemosensory Neurons in the Neurotransmitter-Release Mutant unc-13** *Int J Mol Sci* **25** https://doi.org/10.3390/ijms252312956 | Google Scholar

Bokman E., Pritz C. O., Ruach R., Itskovits E., Sharvit H., Zaslaver A. (2024) **Intricate response dynamics enhances stimulus discrimination in the resource-limited C. elegans chemosensory system** *BMC Biol* **22**:173 https://doi.org/10.1186/s12915-024-01977-z | Google Scholar

Brenner S. (1974) **The genetics of Caenorhabditis elegans** *Genetics* **77**:71–94 https://doi.org/10.1093/genetics/77.1.71 | Google Scholar

Chronis N., Zimmer M., Bargmann C. I. (2007) **Microfluidics for in vivo imaging of neuronal and behavioral activity in Caenorhabditis elegans** *Nature Methods* **4**:727–731 https://doi.org/10.1038/nmeth1075 | Google Scholar

Cook S. J., Jarrell T. A., Brittin C. A., Wang Y., Bloniarz A. E., Yakovlev M. A., Nguyen K. C. Q., Tang L. T., Bayer E. A., Duerr J. S., Bulow H. E., Hobert O., Hall D. H., Emmons S. W. (2019) **Whole-animal connectomes of both Caenorhabditis elegans sexes** *Nature* **571**:63–71 https://doi.org/10.1038/s41586-019-1352-7 | Google Scholar

Dunkel E., Aoki I., Bergs A., Gottschalk A. (2025) **Neurons and molecules involved in noxious light sensation in Caenorhabditis elegans** *G3 (Bethesda)* **15**:jkaf086 https://doi.org/10.1093/g3journal/jkaf086 | Google Scholar

Edwards S. L., Charlie N. K., Milfort M. C., Brown B. S., Gravlin C. N., Knecht J. E., Miller K. G. (2008) **A novel molecular solution for ultraviolet light detection in Caenorhabditis elegans** *Plos Biology* **6**:e198 https://doi.org/10.1371/journal.pbio.0060198 | Google Scholar

Gong J. K., Yuan Y. Y., Ward A., Kang L. J., Zhang B., Wu Z. P., Peng J. M., Feng Z. Y., Liu J. F., Xu X. Z. S. (2016) **The Taste Receptor Homolog LITE-1 Is a Photoreceptor** *Cell* **167**:1252 https://doi.org/10.1016/j.cell.2016.10.053 | Google Scholar

Hanson S. M., Scholuke J., Liewald J., Sharma R., Ruse C., Engel M., Schuler C., Klaus A., Arghittu S., Baumbach F., Seidenthal M., Dill H., Hummer G., Gottschalk A. (2023) **Structure-function analysis suggests that the photoreceptor LITE-1 is a light-activated ion channel** *Current Biology* **33**:3423–3435 https://doi.org/10.1016/j.cub.2023.07.008 | Google Scholar

Hanson S. M., Scholüke J., Liewald J., Sharma R., Ruse C., Engel M., Schüler C., Klaus A., Arghittu S., Baumbach F., Seidenthal M., Dill H., Hummer G., Gottschalk A. (2023) **Structure-function analysis suggests that the photoreceptor LITE-1 is a light-activated ion channel** *Current Biology* **33** https://doi.org/10.1016/j.cub.2023.07.008 | Google Scholar

Itskovits E., Ruach R., Kazakov A., Zaslaver A. (2018) **Concerted pulsatile and graded neural dynamics enables efficient chemotaxis in C. elegans** *Nature Communications* **9**:2866 https://doi.org/10.1038/s41467-018-05151-2 | Google Scholar

Iwanir S., Ruach R., Itskovits E., Pritz C. O., Bokman E., Zaslaver A. (2019) **Irrational behavior in C. elegans arises from asymmetric modulatory effects within single sensory neurons** *Nature Communications* **10**:3202 https://doi.org/10.1038/s41467-019-11163-3 | Google Scholar

Javer A., Currie M., Lee C. W., Hokanson J., Li K., Martineau C. N., Yemini E., Grundy L. J., Li C., Ch'ng Q., Schafer W. R., Nollen E. A. A., Kerr R., Brown A. E. X. (2018) **An open-source platform for analyzing and sharing worm-behavior data** *Nature Methods* **15**:645–646 https://doi.org/10.1038/s41592-018-0112-1 | Google Scholar

Liu J., Ward A., Gao J., Dong Y., Nishio N., Inada H., Kang L., Yu Y., Ma D., Xu T., Mori I., Xie Z., Xu X. Z. (2010) **C. elegans phototransduction requires a G protein-dependent cGMP pathway and a taste receptor homolog** *Nat Neurosci* **13**:715–722 https://doi.org/10.1038/nn.2540 | Google Scholar

Lu W., Zhang J., Huang W., Zhang Z., Jia X., Wang Z., Shi L., Li C., Wolynes P. G., Zheng S. (2024) **DynamicBind: predicting ligand-specific protein-ligand complex structure with a deep equivariant generative model** *Nature Communications* **15**:1071 https://doi.org/10.1038/s41467-024-45461-2 | Google Scholar

McNutt A. T., Francoeur P., Aggarwal R., Masuda T., Meli R., Ragoza M., Sunseri J., Koes D. R. (2021) **GNINA 1.0: molecular docking with deep learning** *J Cheminform* **13**:43 https://doi.org/10.1186/s13321-021-00522-2 | Google Scholar

Montell C. (2009) **A taste of the Drosophila gustatory receptors** *Curr Opin Neurobiol* **19**:345–353 https://doi.org/10.1016/j.conb.2009.07.001 | Google Scholar

Pritz C., Itskovits E., Bokman E., Ruach R., Gritsenko V., Nelken T., Menasherof M., Azulay A., Zaslaver A. (2023) **Principles for coding associative memories in a compact neural network** *eLife* **12** https://doi.org/10.7554/eLife.74434 | Google Scholar

Pu L. J., Wang J., Lu Q. X., Nilsson L., Philbrook A., Pandey A., Zhao L. A., van Schendel R., Koh A., Peres T. V., Hashi W. H., Myint S. L., Williams C., Gilthorpe J. D., Wai S. N., Brown A., Tijsterman

M., Sengupta P., Henriksson J., Chen C. C. (2023) **Dissecting the genetic landscape of GPCR signaling through phenotypic profiling** *Nature Communications* **14**:8410 https://doi.org/10.1038/s41467-023-44177-z | Google Scholar

Richmond J. E., Davis W. S., Jorgensen E. M. (1999) **UNC-13 is required for synaptic vesicle fusion in C. elegans** *Nat Neurosci* **2**:959–964 https://doi.org/10.1038/14755 | Google Scholar

Sengupta P., Chou J. H., Bargmann C. I. (1996) **odr-10 encodes a seven transmembrane domain olfactory receptor required for responses to the odorant diacetyl** *Cell* **84**:899–909 https://doi.org/10.1016/s0092-8674(00)81068-5 | Google Scholar

Taniguchi G., Uozumi T., Kiriyama K., Kamizaki T., Hirotsu T. (2014) **Screening of odor-receptor pairs in Caenorhabditis elegans reveals different receptors for high and low odor concentrations** *Sci Signal* **7**:ra39 https://doi.org/10.1126/scisignal.2005136 | Google Scholar

Toyoshima Y., Tokunaga T., Hirose O., Kanamori M., Teramoto T., Jang M. S., Kuge S., Ishihara T., Yoshida R., Iino Y. (2016) **Accurate Automatic Detection of Densely Distributed Cell Nuclei in 3D Space** *PLoS Comput Biol* **12**:e1004970 https://doi.org/10.1371/journal.pcbi.1004970 | Google Scholar

Ward A., Liu J., Feng Z., Xu X. Z. (2008) **Light-sensitive neurons and channels mediate phototaxis in C. elegans** *Nat Neurosci* **11**:916–922 https://doi.org/10.1038/nn.2155 | Google Scholar

Zhang W., He F., Ronan E. A., Liu H., Gong J., Liu J., Xu X. Z. S. (2020) **Regulation of photosensation by hydrogen peroxide and antioxidants in C. elegans** *PLoS Genet* **16**:e1009257 https://doi.org/10.1371/journal.pgen.1009257 | Google Scholar

Zhang X., Liu J., Pan T., Ward A., Liu J., Xu X. Z. S. (2022) **A cilia-independent function of BBSome mediated by DLK-MAPK signaling in C. elegans photosensation** *Dev Cell* **57**:1545–1557 https://doi.org/10.1016/j.devcel.2022.05.005 | Google Scholar

Zhang Y., Chou J. H., Bradley J., Bargmann C. I., Zinn K. (1997) **The Caenorhabditis elegans seven-transmembrane protein ODR-10 functions as an odorant receptor in mammalian cells** *Proc Natl Acad Sci U S A* **94**:12162–12167 https://doi.org/10.1073/pnas.94.22.12162 | Google Scholar

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**Reviewer #1 (Public review):**

Summary:

This paper describes an interesting phenotype of C. elegans lite-1 mutants. Previous work showed that lite-1 mutants lose a violet/blue light avoidance response. The authors show here that lite-1 mutants also show a defect in negative diacetyl chemotaxis. While wild-type worms avoid diacetyl at high concentrations, lite-1 mutants are instead *attracted* to it. The authors go on to perform Ca2+ imaging in sensory neurons and find that ADL and ASK neurons show altered Ca2+ responses to diacetyl in lite-1 mutants, suggesting LITE-1 is required for these responses. As unc-13 mutants with defective synaptic transmission show similar diacetyl Ca2+ responses as wild-type, this suggests these neurons respond cell autonomously to diacetyl. However, whether lite-1 also acts cell-autonomously is not discussed. Indeed, because unc-13 and lite-1 mutants show different ADL and ASK Ca2+ responses, it seems the diacetyl response regulated by LITE-1 is likely acting outside of those cells. An interesting result that is not commented on is the switching of the valence of the ASK Ca2+ response in lite-1 mutants. ASK neurons still respond to diacetyl, but instead of a strong increase in Ca2+, diacetyl appears to drive it strongly lower. This may be consistent with the switch in valence in the diacetyl chemotaxis assay. It also argues against the idea that LITE-1 is a low-affinity diacetyl receptor that drives avoidance or the Ca2+ responses in ASK, since it is still present in lite-1 mutants. The authors then use a strain that expresses LITE-1 in the body wall muscles and show this expression is sufficient to engender them with sensitivity to diacetyl, as measured through altered swimming and hypercontractility. The authors interpret this result as LITE-1 may act as a diacetyl receptor. The authors test whether a structurally similar molecule, 2,3-pentanedione, shows similar effects, and they find it does. Alpha-fold modeling and molecular docking analysis show where diacetyl might bind to the LITE-1 protein. They then test whether lite-1 mutants show chemotaxis defects to other molecules, as seen with diacetyl. Generally, they find that the observed diacetyl responses are unique, although lite-1 mutants do lose their avoidance response to 2,3-pentanedione. However, unlike the acquisition of diacetyl attraction in lite-1 mutants, 2,3 pentanedione avoidance is *lost*; it is not switched to attraction. Overall, I felt the description of the results and their implications could have been more in-depth. Further, the evidence that LITE-1 is a chemoreceptor itself, rather than acting in some way to shape chemoreceptor responses (via light or otherwise), remains unclear, as conceded by the authors.

Strengths:

Overall, the study follows up on an interesting and useful result. The experiments as presented are generally well-conceived and performed. The authors use a variety of behavioral and imaging approaches to test how LITE-1 mediates diacetyl avoidance.

Weaknesses:

The study is missing experiments needed to resolve whether LITE-1 is doing what they propose. The evidence that LITE-1 is a diacetyl receptor is lacking support since lite-1 mutants have their avoidance and calcium responses flipped, which would not be expected if it were acting solely as an avoidance receptor. Presumably, the authors are concluding that the attractive response that is left in the lite-1 mutant is mediated by ODR-10, but that experiment is not shown. Similarly, the authors concede that "the use of lite-1 point mutants that affect specific LITE-1 function, such as light sensing, channel gating, or binding pocket, could further elucidate LITE-1 mechanisms." This reviewer agrees, and such experiments designed to localize diacetyl binding site(s) would be necessary to conclude definitively that LITE-1 is a diacetyl receptor. The body wall muscle assay used or some other heterologous experimental system could work for such a structure-function analysis. A concern is whether the extensive number of LITE-1 point mutants described in the literature affect cell surface expression vs. receptor function, which might complicate the interpretation of a result showing loss of diacetyl responses.

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**Reviewer #2 (Public review):**

Summary:

Koh and colleagues investigate the broader sensory role of LITE-1, a gustatory receptor previously linked to UV light detection in C. elegans. Their study explores whether LITE-1 also mediates avoidance of specific chemical stimuli-namely, high concentrations of diacetyl and 2,3-pentanedione. They show that LITE-1 is required in the ADL and ASK neurons for calcium responses to diacetyl, and that its expression in body-wall muscles is sufficient to trigger hypercontraction upon odorant exposure. Molecular docking suggests both odorants may directly bind to LITE-1 with micromolar affinity. These findings suggest LITE-1 may act as a multimodal receptor for both light and chemical stimuli.

Strengths:

(1) Methodological Precision: The study is technically strong, with well-executed calcium imaging and quantitative behavioral assays that clearly show neural and muscular responses to chemical stimuli.

(2) Novelty and Scope: The work presents a compelling case for LITE-1 functioning as a multimodal sensor, which is an intriguing expansion of its known role.

(3) Potential Impact: If validated, the findings could significantly advance the understanding of sensory integration in C. elegans, and the tools developed may be broadly useful to the research community.

(4) Relevance to the Field: The study adds to evidence that C. elegans uses non-canonical sensory pathways and may inspire further exploration of multimodal receptor functions in other systems.

Weaknesses:

(1) Lack of Rescue Experiments: The absence of rescue experiments makes it difficult to definitively link the observed phenotypes to loss of lite-1.

(2) Single Loss-of-Function Approach: The reliance on a single genetic mutant limits interpretability. Additional strategies such as RNAi (e.g., neuron-specific knockdown) would provide stronger evidence.

(3) Unclear Neuronal Contribution: While calcium responses in ADL and ASK are reduced, it's unclear which neuron(s) are necessary for behavioral avoidance. Cell-specific rescue or knockdown experiments are needed.

(4) Unvalidated Docking Data: The molecular docking predictions lack experimental validation. Site-directed mutagenesis would be needed to support claims of direct interaction.

(5) Limited Odorant Specificity Testing: Docking analysis does not include non-binding odorants, making it difficult to assess binding specificity.

(6) Incomplete Quantification: Some calcium imaging results (e.g., in AWA neurons of unc-13 mutants) lack statistical comparisons, which limits their interpretive value.

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**Reviewer #3 (Public review):**

In this work, Brown and colleagues report that the photosensor protein LITE-1 of the nematode C. elegans may also be a chemosensor that can be activated by high concentrations of the compound diacetyl. LITE-1 was described as a putative ion channel of the gustatory receptor family, which is mainly constituted by insect odorant receptors. These form tetrameric ion channels that can be activated by odorants. Specificity is achieved by forming heteromeric channels from three copies of the odorant receptor co-receptor (ORCO) and another subunit that resembles ORCO in the pore-forming C-terminus, but brings in a binding site for the respective odorant. LITE-1 has a very similar structure, according to Alphafold3 predictions, and also carries a binding pocket. In LITE-1, this was proposed to be occupied by a light-absorbing molecule that activates the channel when a photon is absorbed. Alternatively, compounds generated by absorption of high-energy photons may be formed in vivo and bound by the LITE-1 binding pocket. Koh et al. now demonstrate that another, non-light-activated compound, diacetyl, at high concentrations, can activate cells expressing LITE-

1. Such (chemosensory) cells are also responsible for the avoidance of high concentrations of diacetyl. LITE-1 activation in excitable cells, i.e, muscles, causes strong body contraction and paralysis, and the authors show that this is also the case when diacetyl is presented. The authors further present molecular docking studies showing that diacetyl could occupy the binding pocket of LITE-1. Last, they show that another compound chemically resembling diacetyl, i.e., 2,3-pentanedione, can also induce avoidance in a LITE-1 dependent manner, though not as potently.

The data are intriguing, and the demonstration of LITE-1 being a diacetyl chemosensor is interesting. Yet, there are a few questions arising that the authors should address.

The authors identified mutants lacking diacetyl responses. In their chemotaxis assay (Figures 1A, B), they show that lite-1 mutants do not avoid high concentrations of diacetyl. However, the animals actually showed attraction, as the chemotaxis index was positive. If the lite-1 animals were insensitive, they should be indifferent, and the chemotaxis index should be close to zero. This means, other neurons contribute to the diacetyl response, and the result of these neurons being activated means/remains attraction? If so, the authors need to rule out any effects of these neurons on the effects they attribute to LITE-1 in the other assays.

The effect of diacetyl on muscle cells (Figure 3C) is pretty rapid, i.e., already during 1 minute after application, the animals are almost maximally contracted. How fast is it really? Can the authors provide a time course with more time points during the first minute? This is a relevant question, as the compound would have to either pass the worm cuticle or enter through the gut and diffuse through the body to reach the muscle cells. Can one expect this to occur within (less than) a minute?

In this context, the authors need to rule out that other mechanisms may be at play. E.g., diacetyl may be immediately sensed by ciliated chemosensory neurons that might release a signaling molecule that leads to activation of LITE-1 in muscles, or that sensitizes it somehow, responding to light used for filming animals. The authors should repeat this assay in a lite-1 mutant background. Furthermore, the authors tested unc-13 mutants to rule out indirect effects on the neurons recorded. Likewise, they should eliminate neuropeptide signaling via unc-31 mutants (a recent paper cited by the authors showed involvement of neuropeptide signaling in LITE-1-mediated light avoidance behavior). Last, to demonstrate that effects are not indirect in response to chemosensory neurons, the authors should repeat the contraction or swimming assay in a tax-4 mutant, which largely lacks chemosensation. This also applies to the chemotaxis assay. Animals should exhibit a chemotaxis index to diacetyl of zero, then.

Does diacetyl activate other neurons expressing LITE-1? A number of cells express LITE-1 at high levels, which the authors have not tested (they restricted their analyses to chemosensory neurons). This is important to address because it leaves the possibility that LITE-1 requires a specific partner only present in these chemosensory neurons to detect diacetyl. This partner would have to be present also in muscles, where diacetyl could activate ectopically expressed LITE-1. According to CeNGEN scRNAseq data, cells expressing LITE-1 can be identified. The ADL and ASH neurons actually come up only at the lowest threshold, so some of the other cells showing much higher levels of LITE-1 mRNAs, i.e., AVG, ALM, PLM, ASG, PHA, PHB, AVM, RIF, or some pharyngeal neurons, should be tested. ASG was among the cells the authors recorded from, but this neuron did not show a response.

The authors need to show that diacetyl responses of ADL and/or ASK can be rescued by expressing LITE-1 specifically in these neurons in a lite-1 mutant background.

Molecular docking studies are not described in detail. How was this done? Diacetyl is a very small molecule. How well can docking algorithms assess this at all? Did the authors preselect the binding pocket, or did the algorithm sample the entire molecular surface of the LITE-1 model and end up with the binding pocket? The latter would be very convincing. The authors should provide control docking experiments with other molecules that caused avoidance in their hands (i.e. benzaldehyde, 2,4,5,trimethlythiazole, isoamyl alcohol, nonanone, octanone), but did not activate LITE-1. Also, they should try docking molecules related to diacetyl, and if there are some that do not dock under the same conditions, such molecules should be used in a behavioral experiment. Ideally, they should also not activate LITE-1. Examples could be, e.g., diacetyl monoxime or 2,4-pentanedione.

Last, the authors should provide a PDB file with the docked diacetyl to allow readers to assess the binding for themselves. Since a large number of mutations of LITE-1 have been reported, it may be that amino acids shown to be essential for LITE-1 function are also required for diacetyl binding. If so, this could be backed up with an experiment.

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